

Post-Quantum Cryptography

Lecture 16 — A Survey of the Families

SEC 595 — Encrypted Computing

- 1 Why Post-Quantum At All
- 2 The Standardisation Process
- 3 Lattice-Based
- 4 Hash-Based
- 5 Code-Based
- 6 Multivariate
- 7 Isogeny-Based
- 8 MPC-in-the-Head
- 9 Comparing the Families
- 10 The Two 2022 Breaks

What Shor's algorithm breaks

A large fault-tolerant quantum computer running Shor's algorithm solves, in polynomial time:

- 1 integer factorisation;
- 2 the discrete logarithm problem in $\mathbb{Z}_p^\times$;
- 3 the discrete logarithm problem on elliptic curves.

That kills RSA, Diffie–Hellman and ECDSA.

And this course

- L8: RSA (factoring), ElGamal (DLog in $\mathbb{Z}_p^\times$), Paillier (factoring $n = pq$).
- L9: BGN, a composite-order pairing group — factoring $n = q_1 q_2$.

All Shor-breakable. The lattice family is why HE has a future.

What Grover's algorithm does

Grover searches an unstructured space of size N in $O(\sqrt{N})$ steps.

Against a λ -bit key it halves the exponent: 2^{128} work becomes about 2^{64} .

The fix: double the key length

AES-256 and SHA-384 are considered post-quantum-adequate; AES-128 is not.

Asymmetric cryptography is destroyed; symmetric cryptography is merely inconvenienced. So every family builds public-key primitives from problems that look nothing like factoring.

Confidentiality is retroactive. Ciphertext recorded today can be decrypted whenever a quantum computer arrives.

Anything that must stay secret for twenty years needs post-quantum protection now.

Authentication is not retroactive: a signature verified today cannot be forged later.

So key establishment migrated first — ML-KEM was in browsers well before signature migration began.

Call issued 2016; first selections July 2022. As of September 2026:

Standard	Scheme	Family	Status
FIPS 203	ML-KEM (Kyber)	lattice	final, 13 Aug 2024
FIPS 204	ML-DSA (Dilithium)	lattice	final, 13 Aug 2024
FIPS 205	SLH-DSA (SPHINCS ⁺)	hash	final, 13 Aug 2024
FIPS 206	FN-DSA (Falcon)	lattice	draft
—	HQC	code	selected 11 Mar 2025

- Three of the five are lattice schemes — the family won.
- SLH-DSA and HQC were chosen *because* it won. One family is a single point of failure.

The additional signatures round

A separate call for signatures *not* based on structured lattices. Fourteen reached round 2; on 14 May 2026, nine advanced to round 3.

Scheme	Family
UOV, MAYO, QR-UOV, SNOVA MQOM, SDitH, FAEST SQIsign HAWK	multivariate MPC-in-the-head isogeny lattice, a different lattice problem

HAWK rests on the lattice *isomorphism* problem, not LWE — so it fails differently.

The family this course has built

L5: γ -SVP, GapSVP, BDD, SIS, q -ary lattices. L6: LWE, its ring and module variants, MSIS.

Scheme	Rests on
ML-KEM	Module-LWE over $R_q = \mathbb{Z}_q[X]/(X^{256} + 1)$, $q = 3329$
ML-DSA	Module-LWE and Module-SIS, $q = 8380417$
FN-DSA	NTRU lattices, Gaussian sampling over a trapdoor
HAWK	the module lattice isomorphism problem

ML-KEM and ML-DSA are the next two lectures (L17, L18).

Why it won, and what it risks

Won on balance

The only family with *simultaneously* small keys, small ciphertexts or signatures, and fast arithmetic (the NTT, L3). Every other family is lopsided.

Risks structure

The reductions are for *unstructured* lattices. ML-KEM uses module lattices over a cyclotomic ring — a tiny, highly symmetric subset.

No known attack exploits that structure at these parameters. Attacks on *ideal*-lattice problems do exist; they do not extend to module lattices only because module lattices are less structured.

Assumption

The hash function is preimage- and collision-resistant. Nothing else.

No number theory, no lattices, no problem invented for the purpose — the most conservative assumption in cryptography.

Lamport one-time signature

Secret random x_0, x_1 ; publish $H(x_0), H(x_1)$. To sign bit b , reveal x_b .

Signing twice reveals both preimages — hence *one-time*. WOTS⁺ compresses this with hash chains.

Merkle tree

Build 2^h one-time keys; authenticate their public keys with a binary hash tree. The root is the long-term public key.

A signature carries the one-time signature plus the h sibling hashes on the path to the root. (Same object as the commitment in L2.)

XMSS and LMS must remember which one-time keys were used — a restored backup can reuse one.

SLH-DSA (SPHINCS⁺)

Removes the state: picks the leaf pseudorandomly from a tree large enough that collisions are negligible. Paid for in signature size.

Scheme	Signature (bytes)
SLH-DSA “small”	7,856
SLH-DSA “fast”	17,088
ML-DSA	2,420
ECDSA	64

Public key: **32 bytes** — the smallest in this lecture. Signing is one to two orders of magnitude slower than ML-DSA.

Not for TLS. Right for firmware and root certificates: signed rarely, verified for decades, where being wrong is unrecoverable.

Hard problem

Given a parity-check matrix $H \in \mathbb{F}_2^{r \times n}$ and a syndrome s , find a low-weight e with $He^T = s$.

NP-hard in general. Information-set decoding is exponential, its exponent essentially unchanged since Prange (1962).

Trapdoor

A code the owner can decode (a binary Goppa code), disguised as a random one.

L2 used Reed–Solomon codes for low-degree testing. Here the sign is flipped: hide an easy code inside a hard-looking one.

The oldest family

McEliece, 1978 — one year after RSA. Never broken.

Scheme	Public key	Ciphertext
Classic McEliece (level 1)	261,120 B	96 B
HQC (level 1)	≈ 2 KB	≈ 4 KB
ML-KEM-512 (for scale)	800 B	768 B

A quarter-megabyte key cannot go in a TLS handshake; a 96-byte ciphertext suits a long-lived static key exchanged once out of band.

NIST chose HQC (random quasi-cyclic codes) for its practical key — paid for with a larger ciphertext and a younger security history.

- 1 SLH-DSA signatures are three times the size of ML-DSA's and signing is far slower. NIST standardised it anyway, on the same day. Give the strongest argument for that decision, and the strongest against.
- 2 Classic McEliece: 261 KB public key, 96-byte ciphertext. Describe a deployment where that is the right trade, and one where it is absurd. What property decides it?

The $\mathcal{MQ}$ problem

Given m quadratic polynomials in n variables over $\mathbb{F}_q$, find a common root.

NP-hard for random systems; Gröbner-basis methods are exponential in general.

Trapdoor

Split the variables into *oil* and *vinegar*; no two oil variables ever multiply.
Fix the vinegar at random: the system is *linear* in the oil.

Secret invertible linear maps disguise the split.

The shape of the trade

- Smallest signatures: UOV signs in about **96 bytes**, comparable to ECDSA.
- Largest public keys: hundreds of kilobytes.
- MAYO and SNOVA try to shrink the key while keeping the signature small.

Four of the nine round-3 signature candidates are multivariate.

An **isogeny** is a structure-preserving map between elliptic curves.

Hard problem

In the supersingular isogeny graph (curves as vertices, small-degree isogenies as edges), recover a path between two given curves.

The graph is an expander; random walks mix fast.

SQIsign (round 3)

About **148-byte** signatures; one to two orders of magnitude slower than ML-DSA at signing.

The family's survivor, after SIDH/SIKE (later in this lecture).

Idea

Take a symmetric primitive (AES, a syndrome-decoding instance, an MQ instance). Prove in zero knowledge that you know the secret key:

- simulate an MPC “in your head”;
- commit to the virtual parties’ views; open a subset;
- remove interaction with Fiat–Shamir.

The transcript is the signature.

- Assumptions as conservative as hash-based; tiny keys (an AES plaintext/ciphertext pair, or a syndrome).
- Signatures of several KB; slow signing.
- FAEST, SDitH, MQOM are in round 3.

Sizes at NIST level 1 (bytes)

Family	Representative	Type	Public key	Ct / sig
Lattice	ML-KEM-512	KEM	800	768
Code	Classic McEliece	KEM	261,120	96
Code	HQC	KEM	$\approx 2,000$	$\approx 4,400$
Lattice	ML-DSA-44	sig	1,312	2,420
Lattice	FN-DSA-512	sig	897	666
Lattice (LIP)	HAWK	sig	$\approx 1,000$	555
Hash	SLH-DSA-128s	sig	32	7,856
Hash	SLH-DSA-128f	sig	32	17,088
Multivariate	UOV	sig	$\approx 10^5$	96
Isogeny	SQIsign	sig	≈ 65	148
MPC-in-head	FAEST	sig	≈ 32	several KB
Classical	ECDSA P-256	sig	64	64
Classical	RSA-2048	sig	256	256

Read by columns. Every family loses in a different corner. ML-DSA is the best at nothing and terrible at nothing — what a default has to be.

The other axis: maturity

Family	Attacked since	Record
Code-based	1978	no break of the original proposal
Multivariate	1988	many breaks, including Rainbow in 2022
Hash-based	1979	reduces to hash security; no break
Lattice	1996	no break at standardised parameters
Isogeny	2006	SIDH/SIKE broken in 2022
MPC-in-head	2007	reduces to symmetric primitives; no break

Time in the arena is evidence, not proof.

A layered oil-and-vinegar signature, and a NIST round-3 *finalist*.

Beullens, 2022

Recovers the level-1 secret key in about a weekend on a laptop.

Rainbow was eliminated.

No quantum computer. A structural weakness in the trapdoor survived five years of public analysis. The scheme had always been insecure; we found out late.

SIDH and its KEM SIKE had by far the smallest keys of any post-quantum scheme.

Castryck–Decru, July 2022

Key recovery in about an hour on one core — weeks after SIKE advanced to round 4.

Maino–Martindale and Robert followed within a month.

The attack used the auxiliary *torsion points* SIDH publishes, via a 1997 theorem of Kani. The isogeny problem itself was not broken.

SQIsign publishes no torsion points.

Both times the hard problem stood; the **trapdoor** leaked.

- ① **Diversity is insurance.** Lattices alone in 2022 would mean no fallback.
- ② **Time in the arena is a security parameter.** Code-based cryptography is forty-eight years old; lattices cannot yet say that.
- ③ **Deploy hybrids.** TLS combines X25519 with ML-KEM-768: secure if *either* holds, for a few hundred bytes.

- ① Both 2022 breaks left the hard problem intact and broke the trapdoor. Does that suggest where to look for the next break? Does the lattice family have an analogous exposed surface?
- ② Hybrid X25519 + ML-KEM-768 costs a few hundred bytes per handshake. When should a hybrid be retired, and who decides?

- 1 Shor breaks factoring and both discrete logs — RSA, ECDSA, and this course's ElGamal, Paillier and BGN. Grover only halves symmetric security.
- 2 Three of the four FIPS standards are lattice schemes; SLH-DSA and HQC are insurance.
- 3 The families trade key size, signature size, speed and assumption age. None wins on all four; lattices lose least.
- 4 Hash-based and MPC-in-the-head assume only symmetric primitives. Code-based is the oldest; multivariate has the smallest signatures and the worst record.
- 5 Rainbow and SIKE fell to classical mathematics in 2022: the trapdoor leaked. So standardise several families and deploy hybrids.